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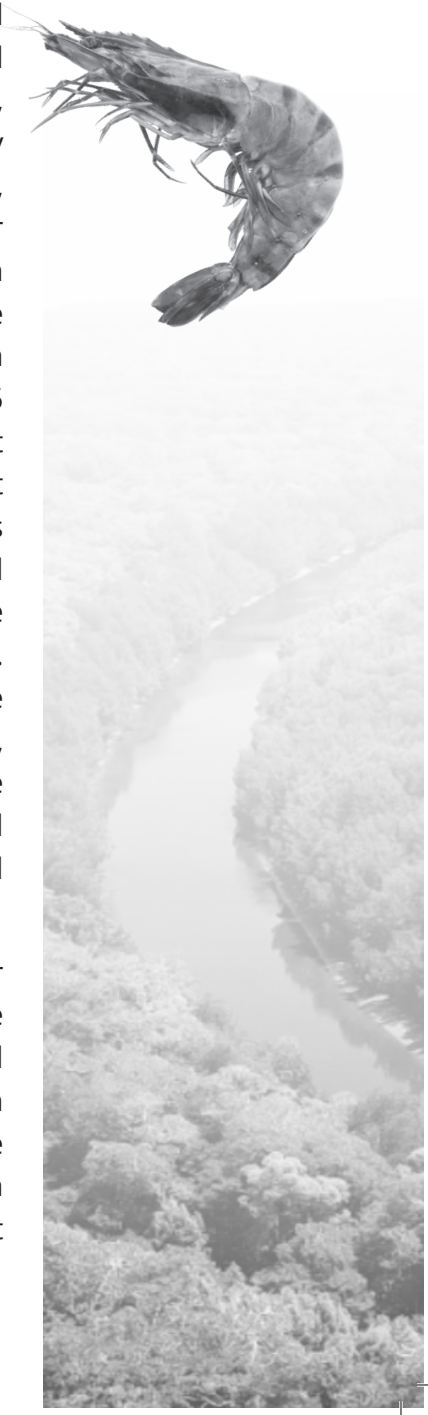
FOREWORD



Mangrove ecosystems are a critical component of the ASEAN region's coastal landscapes, delivering vital ecological, economic, and social services. They provide habitats for diverse species, protect shorelines from erosion, buffer communities against storms, sustain fisheries and livelihoods, and serve as one of the world's most important blue carbon ecosystems. With an estimated 4.5 to 5 million hectares, approximately 30 percent of the global mangrove area, Southeast Asia holds the largest extent of mangroves worldwide. Their conservation and sustainable management are therefore of both regional and global significance. Yet, these ecosystems continue to face pressures from unsustainable exploitation, land conversion, pollution, and climate change, underscoring the urgent need for coordinated and science-based management.

Harvest is a decisive moment for aquaculture farmers, representing the culmination of months of effort and investment. While farmers aim for both high quantity and quality, in practice quantity often takes precedence, even though quality largely determines market

value. To remain competitive, harvesting must be carried out properly and in line with best practices that safeguard both yield and food safety.

The quality of aquaculture products is determined by freshness, safety, nutritional value, and consumer appeal. Freshness is the primary indicator, reflected in neutral sea-like odor, firm texture, natural coloration, and clear eyes in fish. Food safety requires that products remain free from contaminants such as heavy metals, pesticides, and pathogenic bacteria. Nutritional value, including proteins, omega-3, vitamins, and minerals—can only be preserved if cold-chain conditions are maintained. Consumer appeal depends on intact body structure, natural colors, fresh aroma, and overall appearance.

This guideline is aligned with the ASEAN Strategy on Sustainable Mangrove Ecosystem Management, Strategy 4: Implementing appropriate technical intervention and technology in conservation, protection, restoration, and sustainable use of mangrove ecosystems in each AMS. Post-harvest management is a practical technical intervention that connects sustainable aquaculture practices—such as silvofishery—with improved product quality, enhanced value chains, and greater socio-economic benefits for local communities.

Ultimately, this technical guideline is intended as a reference for governments, research institutions, the private sector, and communities. It is expected to not only strengthen understanding of post-harvest practices but also inspire concrete actions that improve aquaculture sustainability, ensure food security, and support regional cooperation in achieving long-term resilience and prosperity.

Author

Indonesia, 2025

I.

INTRODUCTION

A. Background

Harvest is a moment awaited by farmers, as it is the culmination of aquaculture activities undertaken despite numerous challenges. Farmers generally aim to achieve maximum yield, both in terms of quantity and quality. However, in practice, quantity is often prioritized over quality, even though product quality strongly determines market value. To ensure that harvests are both optimal and of high quality, harvesting practices must be carried out properly and in accordance with best standards.

Quality in aquaculture products refers to the overall characteristics of farmed products (such as fish, shrimp, crabs, and others) which determine their freshness, safety for consumption, nutritional value, and appeal to consumers. Among these, freshness is the primary indicator of quality. Freshness is determined by the smell that must be neutral or sea-like odor, the texture is firm and elastic, natural coloration, and clear, convex eyes in fish.

Freshness declines rapidly over time after harvest if products are not immediately cooled or maintained under cold-chain conditions.

Food safety is another essential aspect, ensuring products are free from contaminants such as heavy metals, pesticides, pathogenic bacteria (such as S*almonella*, *Vibrio*), and hazardous chemicals (such as formalin). Food safety is very important to meet national and international food safety standards.

Nutritional value further determines quality of cultivated products, encompassing optimal levels of protein, omega-3, vitamins, and minerals. Improper handling, storage, or a disrupted cold chain can significantly reduce nutritional quality.

The appeal of the aquaculture product must also be maintained, such as the color of the fish must be natural and bright, without pale or darkened gills, surface dryness, or discoloration. The texture of the aquaculture product should be firm and elastic, not soft or excessively slimy. For shrimp and fish products, intact body structure must be essential. The aroma of the cultivated product must remain fresh and sea-like, without unpleasant sour, rotten, or chemical odor, and the taste should remain natural free from bitterness, or chemical taste.

The final quality of aquaculture products is highly dependent on hygienic handling process and cold chain maintained from harvest to consumer, ideally within temperature range of 1-4ºC. At this stage, the enzymatic process and the activity of spoilage bacteria will be inhibited. Collectively, these parameters define the concept of "***Food Quality and Safety***".

This guideline is aligned with the ASEAN strategy on Sustainable Mangrove Ecosystem Management, Strategy 5, which emphasizes the implementation of appropriate technical interventions and technologies in the conservation, protection, restoration, and sustainable use of mangrove ecosystems across ASEAN Member States. More specifically, this technical guideline refers to action plan 6 which highlight opportunities to enhance socio-economic benefits for the local communities. This includes engaging diverse stakeholders in environmentally friendly initiatives such as silvofishery, eco-tourism, development of derivative food products, and improvement of packaging, marketing, promoting, branding, and off-taking schemes.

B. Objective

This Technical Guideline for Post-Harvest Management of Fishery Products aims to improve the quality of post-harvest aquaculture products in order to increase the selling value of aquaculture products and maintain food quality and safety.

C. Scope

The scope of post-harvest guidelines includes:

1. The process of harvesting fish from ponds properly and correctly
2. Traceability

D. Operational Definition

Food Quality and Safety (FQS)	:	Important aspects of product quality are related to food safety and quality.
Hatchery	:	A place for breeding shrimp/fish fry, before finally being cultivated in ponds.

Monoculture	:	An aquaculture system that only cultivate one type of organism in one pond (e.g., only shrimp or only milkfish).
Fish Processing Plant	:	A facility where fish and other aquatic products are prepared for human consumption or further processing. It serves as a critical link between aquaculture fisheries and the market
Harvest	:	The activity of taking the results of aquaculture after reaching the appropriate size for marketing. Harvesting is the final process of aquaculture.
Polyculture	:	An aquaculture system that cultivates several types of organisms in one pond (e.g. shrimp and milk fish).

II.

BEST PRACTICES OF HARVEST PROCESS IN AQUACULTURE

Post-harvest handling plays a critical role in determining the quality of aquaculture products, which in turn directly affects the value of the product. In order to ensure high value of aquaculture products, the post-harvest handling process adhere to the principles of maintaining quality and safety, collectively referred to as "Food Quality and Safety" (FQS). The following section outlines proper harvesting methods to safeguard FOS In aquaculture products.

A. Harvest Time

Harvest time is greatly influenced by several factors, such as the type of fish (fish or shrimp), pond size, water quality, and desired harvest size.

For example:

1. Milkfish (*Chanos chanos*), under normal conditions, is usually harvested when it reaches a weight of 250-500 g/fish (size of 2-4 fish/kg). Generally, milkfish have reached this size after being cultivated for 4-6 months.
2. White leg Shrimp (*Litopenaes monodon*) has a good selling value if it has reached a weight of around 10 g/shrimp (size of 100 shrimp/kg) in a 2-month cultivation period.
3. Tiger prawns (*Penaeus monodon*) can also be harvested at a size of 80-100 g/shrimps with a cultivation period of 5-6 months.
4. Seaweed harvested after approximately 45 days.
5. Mud crab (*Scylla* spp.) harvested after 30–60 days depending on growth and market demand.

Harvesting can also be done outside the above conditions, due to markets fluctuation (e.g. price considerations) in household financial needs. Farmers may also practice “partial harvesting”, explained further in Section D.

Total harvest is usually conducted during low tide. Partial harvest, by contrast, can be done at any time when the fish/shrimp have entered market size and ideally carried out in the morning, afternoon, or evening, during calm weather with ambient temperature between 20–25. Avoiding excessive heat is important, as high temperature can compromise product quality.

B. Harvest Preparation

Prior to harvesting, there are several preparations that need to be done, including the following:

1. Ensure that the harvest area around the pond (to temporarily store the harvested fish) is free from trash and sharp objects to avoid damage to the fish during harvest.
2. Prepare and ensure the cleanliness of the tools and materials needed for harvesting, such as nets, baskets, scales, and containers (fiber tubs) to store fish and crushed ice.
3. Use harvesting tools that have a smooth surface (not sharp) and are still in good condition (not weathered) to avoid physical damage and physical contamination of the shrimp.
4. Prepare several containers to sort and separate the shrimp: group high-quality shrimp by size, and place low-quality or below-standard (BS) shrimp such as those with soft shells due to molting into a separate container.
5. Prepare clean water to clean the harvest from dirt and mud.

C. Partial Harvest

Partial harvest revers to the collection of a portion of cultured fish or shrimp while cultivation cycle is still ongoing. This method is practiced when some individuals have reached market size or for other specific purposes.

1. Partial harvest of fish/shrimp can be performed without draining the pond water. A seine net (arad) may be drawn across a specific section of the pond, as illustrated in Figure 1.

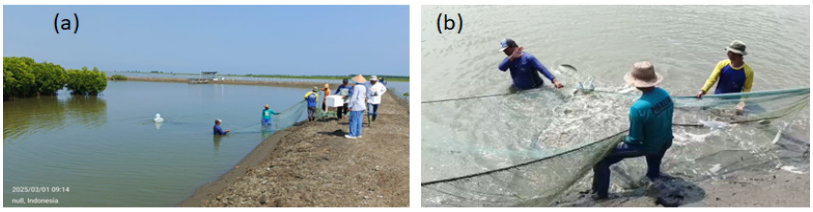


Figure 1. a,b. Partial harvest of milkfish with "arrad" net

2. Shrimp partial harvesting, whether cultured in polyculture with fish or in monoculture systems, can be conducted using traditional bamboo traps known as *bubu*, as shown in Figure 2. The *bubu* are typically set in the afternoon (around 16:00–17:00) and collected the following morning (around 06:00–07:00). Each *bubu* can yield approximately 2–5 kg of shrimp overnight, depending on population density in the pond.

Figure 2. "Bubu" fishing gear

D. Total Harvest

1. Total harvest is recommended during low tide, allowing pond water to be drained by gravity.
2. Install a 'condom net' (a collection net placed at the outlet get that can hold fish/shrimp) to ensure that all outgoing water from the pond passes through it.

3. The mesh size must be appropriate to hold both shrimp and fish. The length of the net should be adjusted to pond size. For larger the pond, a longer the net (> 10 - 30 meters) is required to reduce water pressure on the captured shrimp or fish (Figure 3).
4. Once the net is properly installed, gradually open the water gate by lifting the boards one at a time, in accordance with the strength of the water flow.
5. As the water flows out, the shrimp and fish will be carried away and collected in the net.

Figure 3. a,b. Nets for catching fish in traditional ponds.

6. Collect the trapped fish/shrimp from the condom net and transfer them to the designated harvest area.
7. Separate fish, shrimp, and other species into different containers (Figure 4).

Figure 4. a,b,c. Sorting fish and shrimp immediately after being lifted from the pond.

8. To accelerate the harvesting process, a drag net can also be used simultaneously inside the pond.
9. Rinse the harvested fish and shrimp with clean water to remove mud and debris.

10. Immediately place the fish/shrimp into insulated boxes with crushed ice as soon as possible (Figure 5). Freshness is best maintained at temperatures below 4°C.

Figure 5. a,b. Shrimp and fish are immediately put into a box filled with ice

E. Post Harvest Handling

1. The most crucial stage of the harvest process is weighing. To preserve the quality of shrimp and fish, it is essential to maintain the cold chain throughout handling. Ideally, shrimp and fish should be allowed to die under cold conditions, with their core temperature reduced to approximately 1-4°C (Figure 6). This indicates that shrimp and fish must be placed into ice slurry immediately after removal from the pond, while still alive.
2. A common weakness in current practice is allowing shrimp and fish to die in open air, often under direct sunlight, followed by draining to remove excess water (Figure 6a). Intermediate buyers (collectors) typically prefer not to weigh shrimp or fish when they still contain excess water or ice, due to concerns about fluctuations in weight and size. The weight and size of aquaculture products, particularly shrimp, may increase after being soaked in an ice slurry. For this reason, weighing is usually carried out before the shrimp or fish are placed into ice, with an additional draining step to ensure no water drips from their bodies. The increase in shrimp weight after immersion in ice slurry can reach 8-12%, depending on the storage duration or transportation time from pond

to the Fish Processing Unit. While this practice may provide short-term benefits to collectors or intermediary traders, it negatively impacts on the overall product quality.

Figure 6. (a) Mishandled post-harvest shrimp (b) Good handling (c) Measurement of the central temperature of the shrimp

3. It is therefore essential that fish or shrimp are placed into containers with ice immediately after removing it from the pond.
4. Proper packaging requires placing fish/shrimp in a fiber container, beginning with a base layer of ice, followed by a layer of fish/shrimp, and then another layer of ice, until the container is full. The ratio of ice to fish/ shrimp should be at least 1:1.
5. Transport the containers to the processing plant as quickly as possible.
6. The process of shrimp death at room temperature >20 the rigor mortis process will accelerate, reducing the quality of the flesh. When cooked (steamed) many shrimps may develop "**mushy flesh"** or **"soft, disintegrated flesh"** which diminish their natural taste and freshness.

NOTE:

The method for determining the final weight of harvested shrimp (which must be paid by the buyer) in relation to the soaking process in ice, known to increase shrimp weight, is set as follows:

1. An agreement must first be established between the buyer (intermediary trader, factory, or processing plant) and the farmer. One method that has been applied in an integrated

shrimp farming company in Lampung, Indonesia (2003 – 2012) is described below, for shrimp commodities. Prepare "ice slurry" by mixing crushed ice (cubes) with water in a ratio of 2 bags of ice to 1-1.5 parts water.

2. Take a shrimp sample of 3-5 kg, drain until no water drips, then weigh it (recorded as "a" kg)
3. Place the shrimp into the "ice slurry" with a minimum ratio of shrimp to "ice slurry" of at least 1:1.
4. Leave it for a while, then measure the core-temperature of the shrimp (Figure 6 c) until the temperature drops to <4°C. Keep the shrimp in the "ice slurry" for 48 - 72 hours, then drain and weigh (recorded as "b" kg). Typically, the weight increases by 8-12%, depending on the soaking duration.
5. The soaking duration reflects the transportation time from pond to factory (processing plant) which usually takes 2-3 days.
6. Calculate the weight difference between the shrimp before soaking ("a" kg) and after soaking ("b" kg).

$$\%\, weight\ gain = \frac{a-b}{a}\ x100\%$$

Example = c%

Thus, if the actual harvest is 750 kg (farmed), then after soaking during transportation, the weight will increase by

750 kg x "Y"% = "c" % .

where "Y" usually varies between 8 – 12%.

EXAMPLE:

1. The harvest is placed directly into the ice slurry, then immediately packed and transported to the factory (processing plant).
2. Upon arrival at the Factory, the shrimp are weighed, with a total of 10 tons.
3. Therefore, the estimated weight of the shrimp before being placed into the ice slurry (immediately after harvest) is:
4. 10 tons – (12% x 10 tons) = 8.8 tons.

Hence, the amount of shrimp before soaking ice (immediately after harvest) is 8.8 tons. This is the amount that must be paid by the buyer.

III.

TRACEABILITY

Every aquaculture product originated from ponds must be registered with comprehensive records documenting all treatments, conditions, or events from pond preparation, fry/juvenile stocking, cultivation conditions of fish/shrimp to harvest procedures. These records should ideally be managed through a digital information system to ensure traceability, allowing the full production history to be accessed at any time. Traceability is one of the key requirements in implementation of Good Aquaculture Practices (GAP), which is currently being promoted by both international aquaculture certification bodies and the Government of Indonesia. It has also become an essential prerequisite for products entering export markets.

For aquaculture products, traceability is a shared responsibility among farmers, intermediary traders or brokers, and processing plants (factory). However, processing plants serve as the frontline of traceability management, as they bear ultimate responsibility for providing

verified product information to buyers, particularly in export destination countries. Therefore, processing plants must take a proactive role in promoting and ensuring compliance with proper documentation practices.

Traceability in the hatchery is a record that contains complete information on shrimp seed production, including:

1. Pond condition and cultivation methodology (traditional, semi-intensive or intensive).
2. Broodstock information: origin, health status (Specific Pathogen Free-SPF), number of seeds, classification, and separation of healthy/unhealthy fry.
3. Production process: method, quality of fry produced, and socking schedule.
4. Monitoring of shrimp health during cultivation, including treatments or therapeutants administered if disease symptoms appear.
5. Treatment applied in hatchery (use of fertilizer, lime, pesticides, and others)
6. Purchase transactions of broodstock and product inputs.
7. Each batch of fry release must be accompanied by a logbook containing the above-mentioned records.

Note: only disease-free fry may be marketed.

Traceability at pond level should include the following information:

1. Pond owner's name and address.
2. Seed information
3. Materials used during pond and water preparation.
4. Cultivation process:

- » Cultivation method applied: traditional, semi-intensive, or intensive.
- » Materials used during cultivation: feed, fertilizer (organic/inorganic), chemicals used, and their dosage applied for maintaining shrimp health.

5. Harvest information: date, tools and materials used during harvest, and product quality of shrimp at harvest and upon arrival in the processing plant.

Traceability at the factory level, at least includes the following information:

1. Origin of shrimp: farmer name, address, and harvest date
2. Shrimp quality/health: whether the harvest was normal or emergency (due to disease).
3. Production process: Harvest date, processed date, and date of release from the plant.
4. Shrimp purchase transactions and use of other production inputs.
5. Each product leaving the factory must carry a unique traceability code or barcode containing the above information.

Traceability is a shared responsibility between farmers, intermediary traders, and Processing Plant (Factory), which must be coordinated and verified by the processing plant to ensure compliance with national and international standards.